

School Computer Network Design and Simulation

1. Network Objective

The objective of this project is to design and simulate a computer network for a school using Cisco Packet Tracer. The network provides communication between the administration, educational laboratories, and wireless users, as well as access to shared network resources.

2. Number of Users and Devices (Sample)

The network includes:

- 24 personal computers
- 2 network printers
- 1 server
- 2 laptops
- 1 smartphone
- 1 Access Point

Total: 31 devices

3. Selected Network Topology

A Star Topology is used in the network, where end devices are connected to central switches. This topology simplifies network management and troubleshooting.

4. Devices Used

- 2 Routers (Cisco ISR4321)
- 2 Switches (Cisco 2960)
- 24 PCs
- 2 Printers
- 1 Server
- 1 Access Point
- 2 Laptops
- 1 Smartphone

5. IP Addressing Plan

The network uses three separate IPv4 networks:

- 192.168.1.0/25 — VLAN 10
- 192.168.3.0/25 — VLAN 20
- 192.168.2.0/30 — Router-to-router connection

VLAN 10 – Educational Section

Network: 192.168.1.0/25

- Gateway: 192.168.1.1
- Subnet Mask: 255.255.255.128
- DHCP Address Range: 192.168.1.2 – 192.168.1.126

Computers assigned to VLAN 10 receive their IP addresses automatically through the DHCP service.

VLAN 20 – Administration

Network: 192.168.3.0/25

- Gateway: 192.168.3.1
- Subnet Mask: 255.255.255.128

The devices assigned to the administration network use statically configured IP addresses:

- Server: 192.168.3.2
- Printer 1: 192.168.3.3
- Printer 2: 192.168.3.4

Router-to-Router Connection

A separate 192.168.2.0/30 network is used for the connection between the routers.

Subnet Mask:

255.255.255.252

A /30 subnet provides two usable host addresses, which is sufficient for connecting two routers.

6. VLAN Configuration

Two Virtual Local Area Networks (VLANs) were created to improve network segmentation and management:

- **VLAN 10** — Educational Section
- **VLAN 20** — Administration

Inter-VLAN communication is implemented using the **Router-on-a-Stick** method.

7. DHCP Service

DHCP is configured for VLAN 10. As a result, the corresponding computers automatically receive:

- IP address
- Subnet mask
- Default gateway

8. Routing

The network uses two routers connected through a serial connection.

Dynamic routing between the different networks is implemented using the **RIP** routing protocol.

9. Network Security

The following basic security mechanisms are configured to protect network devices:

- **Console Password** — protects local access to routers and switches.
- **Enable Secret Password** — protects privileged EXEC mode.
- **VTY Password** — controls remote administrative access.
- **Service Password Encryption** — protects passwords stored on network devices.

The network also uses:

- VLAN-based network segmentation
- Separation of the administration and educational networks

10. Testing

The network functionality was tested using the `ping` command.

The testing confirmed that:

- Devices within VLAN 10 can communicate successfully.
- Devices within VLAN 20 can communicate successfully.
- Communication between different VLANs is successful.
- The connection between the routers works correctly.
- DHCP automatically assigns IP addresses to VLAN 10 users.

11. Estimated Budget

Device	Quantity	Unit Price (USD)	Total Price (USD)
Cisco ISR 4321 Router	2	\$769.00	\$1,538.00
Cisco Catalyst 2960 Switch (24-port)	2	\$150.00	\$300.00
Standard Office PC (i5, 16GB RAM, 512GB SSD)	24	\$500.00	\$12,000.00
Office Laptop (i5, 8GB RAM)	2	\$450.00	\$900.00
Network Printer (Printer-PT)	2	\$180.00	\$360.00
Server (Server-PT)	1	\$1,000.00	\$1,000.00
Wi-Fi Access Point	1	\$150.00	\$150.00
Cisco Serial Cable	1	\$5.00	\$5.00
Copper Straight-Through Cable (Cat6, 3m)	34	\$0.75	\$25.50
Total			\$16,278.50

Conclusion

This project represents a medium-sized school computer network model that provides network segmentation, automatic IP address assignment through DHCP, routing between different networks, and basic network security mechanisms.

The network was designed and simulated using Cisco Packet Tracer.